Day 6 — HTML Fundamentals

Simple Questions, Answers, Examples, and Practice Webpages

1. What is HTML?

HTML stands for HyperText Markup Language. It defines the structure and content of a webpage, such as headings, paragraphs, links, images, tables, and forms. HTML is a markup language, not a programming language.

2. What is the basic HTML document structure?

A standard HTML page uses a doctype, an html element, a head section for page information, and a body section for visible content.

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>My First Page</title>
</head>
<body>
    <h1>Hello, World!</h1>
    <p>This is my first webpage.</p>
</body>
</html>
```

3. What are headings and paragraphs?

Headings use h1 through h6, from the most important heading to the least important. The p element is used for a paragraph of text.

```
<h1>My Profile</h1>
<h2>About Me</h2>
<p>I am learning web development.</p>
```

4. How do links and images work?

The a element creates a hyperlink using href. The img element displays an image using src and should include meaningful alt text.

```
<a href="https://example.com">Visit Example</a>
<img src="profile.jpg" alt="Profile photo" width="180">
```

5. How do you create lists and tables?

Use ul for a bulleted list, ol for a numbered list, and li for each list item. Use table with tr rows, th header cells, and td data cells to display tabular information.

```
<ul>
    <li>HTML</li>
    <li>CSS</li>
</ul>

<table>
    <tr><th>ID</th><th>Name</th><th>Marks</th></tr>
    <tr><td>101</td><td>Asha</td><td>88</td></tr>
    <tr><td>102</td><td>Ravi</td><td>76</td></tr>
</table>
```

6. What are forms and input fields?

Forms collect information from users. Common input types include text, email, password, number, date, radio, and checkbox. Labels help users and assistive technologies understand each field. HTML validation attributes such as required and min can help check input.

```
<form action="#" method="post">
    <label for="name">Full name:</label>
    <input type="text" id="name" name="name" required>

    <label for="email">Email:</label>
    <input type="email" id="email" name="email" required>

    <button type="submit">Register</button>
</form>
```

7. What is semantic HTML?

Semantic HTML uses elements that describe the purpose of their content. Examples include header, nav, main, section, article, aside, and footer. Semantic elements improve page structure, accessibility, and maintainability.

```
<header><h1>My Website</h1></header>
<nav><a href="#about">About</a></nav>
<main>
    <section id="about">
        <h2>About Me</h2>
        <p>Welcome to my profile.</p>
    </section>
</main>
<footer><p>Copyright 2026</p></footer>
```

8. Practice Task: Personal Profile Webpage

Create a file named profile.html and save the following example. Replace the sample name and details with your own.

```
<!DOCTYPE html>
<html lang="en">
```

```
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>My Profile</title>
</head>
<body>
    <header>
        <h1>Your Name</h1>
        <p>Student | Aspiring Web Developer</p>
    </header>

    <nav>
        <a href="#about">About</a> |
        <a href="#skills">Skills</a> |
        <a href="#contact">Contact</a>
    </nav>

    <main>
        <section id="about">
            <h2>About Me</h2>
            <p>I am learning HTML and web development.</p>
            <img src="profile.jpg" alt="My profile photo" width="150">
        </section>

        <section id="skills">
            <h2>My Skills</h2>
            <ul>
                <li>HTML</li>
                <li>CSS</li>
                <li>Java</li>
            </ul>
        </section>

        <section id="contact">
            <h2>Contact</h2>
            <p>Email: student@example.com</p>
        </section>
    </main>

    <footer><p>My Profile Page</p></footer>
</body>
</html>
```

9. Practice Task: Student Registration Form

This form demonstrates common input types. A real application needs a backend to receive and store submitted data; action="#" is only a placeholder.

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Student Registration</title>
</head>
<body>
    <main>
        <h1>Student Registration Form</h1>
        <form action="#" method="post">
            <label for="name">Full Name:</label>
            <input type="text" id="name" name="name" required><br><br>

            <label for="email">Email:</label>
            <input type="email" id="email" name="email" required><br><br>

            <label for="age">Age:</label>
            <input type="number" id="age" name="age" min="1" max="120"><br><br>

            <label for="dob">Date of Birth:</label>
            <input type="date" id="dob" name="dob"><br><br>

            <label for="course">Course:</label>
            <select id="course" name="course">
                <option value="bsc">BSc</option>
                <option value="mca">MCA</option>
                <option value="other">Other</option>
            </select><br><br>

            <button type="submit">Submit</button>
            <button type="reset">Reset</button>
        </form>
    </main>
</body>
</html>
```

10. Task: Student Records Table

```
<table>
    <caption>Student Records</caption>
    <thead>
        <tr><th scope="col">ID</th><th scope="col">Name</th><th scope="col">Course</th><th scope="col">Marks</th></tr>
    </thead>
    <tbody>
        <tr><td>101</td><td>Asha</td><td>BSc</td><td>88</td></tr>
        <tr><td>102</td><td>Ravi</td><td>MCA</td><td>76</td></tr>
```

```
        <tr><td>103</td><td>Meera</td><td>BSc</td><td>91</td></tr>
    </tbody>
</table>
```

Day 6 — Task Checklist

■ Create and open an HTML file in a browser.

■ Build a personal profile page using semantic elements.

■ Add headings, paragraphs, links, an image, and a list.

■ Create a student registration form with labels and suitable input types.

■ Create a student records table using thead and tbody.

Quick Revision

HTML: webpage structure. **Headings:** h1–h6. **Paragraph:** p. **Link:** a. **Image:** img with alt text. **Lists:** ul/ol/li. **Table:** table/tr/th/td. **Forms:** form/label/input/button. **Semantic HTML:** header/nav/main/section/article/footer.